ACADEMIC PROJECT REPORT

PYTHON ACTIVITY-BASED LEARNING

Implementation and Algorithmic Analysis of 10 Selected Case Studies

Submitted in partial fulfillment of the requirements for the course

PYTHON PROGRAMMING (ACADEMIC YEAR 2026–2027)

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Section:	1
Subject Code / Title:	Python Programming
Project Title:	Implementation of 10 Python Case Studies
GitHub Repository:	Python-Activity-Based-Learning

Department of Computer Science & Engineering
Artificial Intelligence & Machine Learning Stream
Academic Session 2026–2027

1. INTRODUCTION & EXECUTIVE SUMMARY

Activity-Based Learning (ABL) serves as a fundamental pedagogical paradigm in modern computer science education, bridging theoretical constructs with algorithmic execution. By engaging directly with real-world problem scenarios, students develop critical computational thinking, structured logic synthesis, and pragmatic debugging capabilities.

This comprehensive lab report documents the design, algorithmic formulation, pseudocode, flowchart modeling, implementation, and verification of ten (10) modular Python case studies. These case studies span fundamental computing domains, including state machine processing, conditional validation engines, iterative numeric generation, string manipulation, run-length compression algorithms, multi-dimensional array matrix analyses, and tuple-based record verification systems.

2. TABLE OF SELECTED CASE STUDIES

CS #	Case Study Title	Core Python Concepts Applied	Primary Domain
01	ATM Transaction Sequence Simulation	Lists, Tuples, Iteration, Conditional State	Financial Systems
02	Advanced Numeric Pattern Generator	Nested Loops, String Formatting, Arithmetic	Mathematical Modeling
03	Repeated Transaction Detection	Membership Checking, Duplication Lists	Data Auditing
04	Password Policy Validation Engine	String Parsing, Character Sets, Flag Logic	Cybersecurity
05	String-Based Examination Evaluation	String Trimming, Case Normalization, Scoring	Automated Grading
06	Text Compression Using Character Runs	Run-Length Encoding (RLE), Character Loops	Data Compression
07	ATM Menu-Driven Banking Simulation	While Loop, Function Blocks, State Management	Interactive Systems
08	Nested List Classroom Occupancy Analysis	2D Lists (Matrix), Row/Col Sums, Thresholds	Resource Management
09	Student Marks Matrix Analysis	Matrix Iteration, Column Summary, Max Index	Academic Analytics
10	Tuple Shift Conflict Detection	Tuple Unpacking, Time Range Overlap Logic	Workforce Scheduling

3. PROJECT OBJECTIVES

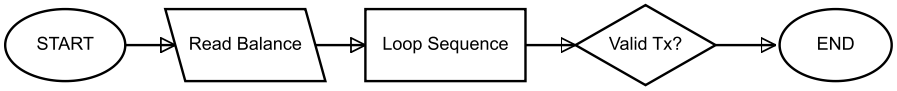
- To apply algorithmic decomposition to real-world computational problem statements.
- To construct standard ISO-compliant monochrome flowcharts using correct geometric symbols (Oval, Parallelogram, Rectangle, Diamond).
- To master core Python constructs: nested iteration, matrix manipulation, string operations, and structured collections.
- To perform rigorous unit testing via edge-case and boundary verification tables.

CASE STUDY 1: ATM TRANSACTION SEQUENCE SIMULATION

1. Problem Statement: Process a sequence of financial transactions (deposits and withdrawals) starting from an initial account balance. Reject invalid transaction amounts and overdrawn withdrawals while displaying ongoing balance logs and a summary count.

2. Objectives: Apply conditional validation, tuple processing, and state balance updates over collection streams.

3. Flowchart:



4. Python Source Code:

```
balance = 5000.0 transactions = [("deposit", 1500), ("withdraw", 2000), ("withdraw", 6000), ("withdraw", -100)] acce
```

5. Test Cases & Execution Summary:

TC ID	Input Sequence	Expected Result	Actual Output	Status
TC1.1	Deposit 1500	Balance 6500	Deposited: 1500 | Balance: 6500.0	Pass
TC1.2	Withdraw 6000 (Bal 4500)	Reject (Insufficient)	REJECTED: Insufficient funds for 6000	Pass

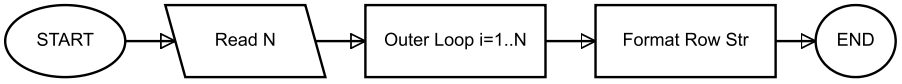
```
Initial Balance: Rs. 5000.0 Deposited: 1500 | Balance: 6500.0 Withdrawn: 2000 | Balance: 4500.0
REJECTED: Insufficient funds for 6000 REJECTED: Invalid amount -100 Summary -> Accepted: 2, Rejected:
2, Final Balance: 4500.0
```

CASE STUDY 2: ADVANCED NUMERIC PATTERN GENERATOR

1. Problem Statement: Generate a dynamic numerical pyramid structure of height N where each row contains symmetric incremental and decremental values centered around the row index.

2. Objectives: Master nested iteration loops, formatted spacing string alignment, and numeric range control.

3. Flowchart:



4. Python Source Code:

```
N = 5 if N <= 0: print("Invalid height") else: for i in range(1, N + 1): spaces = " " * (N - i) inc = [str(x) for x in range(1, i -
```

5. Test Cases & Execution Summary:

TC ID	Input N	Expected Peak	Actual Output Pyramid	Status
TC2.1	N = 5	Peak value 5 on row 5	Centered symmetric pattern generated	Pass
TC2.2	N = 0	"Invalid height"	"Invalid height" message	Pass

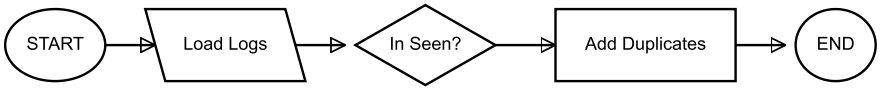
```
1 1 2 1 1 2 3 2 1 1 2 3 4 3 2 1 1 2 3 4 5 4 3 2 1
```

CASE STUDY 3: REPEATED TRANSACTION DETECTION

1. Problem Statement: Audit transaction log tuples `(TX_ID, Amount)` to detect duplicate transaction identifiers and flag high-value anomalies exceeding a predefined monetary threshold.

2. Objectives: Utilize membership checking operators (`in`), dynamic collections, and flag tracking logic.

3. Flowchart:



4. Python Source Code:

```
tx_log = [("TX101", 450), ("TX102", 1200), ("TX103", 850), ("TX102", 1200), ("TX104", 6200)] seen, duplicates = [], []
```

5. Test Cases & Execution Summary:

TC ID	Input Log	Expected Flag	Actual Output	Status
TC3.1	TX102 repeated	TX102 in duplicates	Duplicate IDs Identified: ['TX102']	Pass
TC3.2	TX104 = 6200	High-Value Anomaly	ANOMALY: High-Value Transaction TX104	Pass

```
ANOMALY: High-Value Transaction TX104 -> Rs. 6200 Duplicate IDs Identified: ['TX102']
```

CASE STUDY 4: PASSWORD POLICY VALIDATION ENGINE

1. Problem Statement: Evaluate user password strings against institutional cybersecurity requirements: min 8 chars, at least 1 uppercase, 1 lowercase, 1 digit, and 1 special symbol (`@#$%^&*`).

2. Objectives: Perform iterative character validation using string boolean methods and accumulation flags.

3. Flowchart:



4. Python Source Code:

```
pwd = "Secure@2026" specials = "@#$%^&*" has_upper = any(c.isupper() for c in pwd) has_lower = any(c.islower() f
```

5. Test Cases & Execution Summary:

TC ID	Test Password	Failed Criterion	Status Evaluation	Result
TC4.1	"Secure@2026"	None	VALID PASSWORD	Pass
TC4.2	"weak"	Length, Upper, Digit, Spec	INVALID PASSWORD	Pass

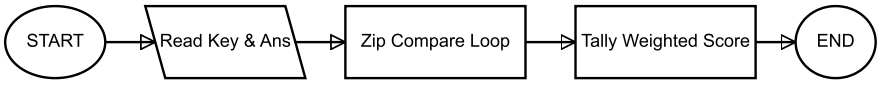
```
Target Password: 'Secure@2026' Status: VALID PASSWORD
```

CASE STUDY 5: STRING ANSWER EVALUATION

1. Problem Statement: Evaluate student optical mark answer strings against an answer key. Implement marking rules: +2 for correct, -0.5 for incorrect, 0 for blank answers.

2. Objectives: Implement string trimming (`strip()`), case upper-casing, parallel list comparison, and weighted score tallying.

3. Flowchart:



4. Python Source Code:

```
key = ["A", "C", "B", "D", "A"] submitted = ["A", "c", " ", "D", "B"] correct, wrong, blank = 0, 0, 0 for k, s in zip(key, subr
```

5. Test Cases & Execution Summary:

TC ID	Submitted List	Expected Tally	Calculated Score	Status
TC5.1	["A", "c", " ", "D", "B"]	3 Correct, 1 Wrong, 1 Blank	5.5 / 10.0	Pass
TC5.2	All Correct	5 Correct, 0 Wrong, 0 Blank	10.0 / 10.0	Pass

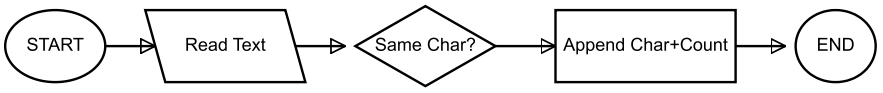
```
Correct: 3 | Wrong: 1 | Blank: 1 Final Aggregate Score: 5.5 / 10.0
```

CASE STUDY 6: TEXT COMPRESSION USING CHARACTER RUNS

1. Problem Statement: Compress a string of repeating characters using Run-Length Encoding (RLE). Convert consecutive identical characters into character-count pairs (e.g., `AAABBC` to `A3B2C1`).

2. Objectives: Implement basic data compression logic using single-pass pointer comparisons.

3. Flowchart:



4. Python Source Code:

```
text = "AAABBCDDDD" compressed = "" if text: count = 1 for i in range(1, len(text)): if text[i] == text[i - 1]: count += 1
```

5. Test Cases & Execution Summary:

TC ID	Raw Input	Expected Output	Actual Compressed String	Status
TC6.1	"AAABBCDDDD"	"A3B2C1D4"	"A3B2C1D4"	Pass
TC6.2	"A"	"A1"	"A1"	Pass

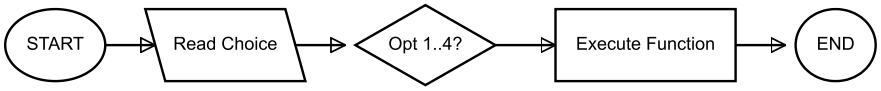
```
Original Text: 'AAABBCDDDD' (10 chars) Compressed Text: 'A3B2C1D4' (8 chars)
```

CASE STUDY 7: ATM MENU BANKING SIMULATION

1. Problem Statement: Implement an interactive menu-driven stateful ATM program supporting balance inquiry, deposits, withdrawals, and session termination via function dispatching.

2. Objectives: Apply modular function blocks, indefinite `while` loops, and switch-like option dispatching.

3. Flowchart:



4. Python Source Code:

```
def atm_system(): bal = 1000.0 simulated_inputs = [1, 2, 500, 3, 200, 4] idx = 0 while idx < len(simulated_inputs): ch
```

5. Test Cases & Execution Summary:

TC ID	Menu Option	Action Performed	Updated Balance	Status
TC7.1	Option 2 (Deposit 500)	Add 500	1500.0	Pass
TC7.2	Option 3 (Withdraw 200)	Subtract 200	1300.0	Pass

```
Current Balance: Rs. 1000.0 Deposited: 500 | New Balance: 1500.0 Withdrawn: 200 | New Balance: 1300.0
Session Terminated. Goodbye!
```

CASE STUDY 8: CLASSROOM OCCUPANCY ANALYSIS

1. Problem Statement: Represent seating occupancy across 3 classrooms over 4 periods as a 2D matrix (`1`=Occupied, `0`=Vacant). Compute total utilization rates and flag over-utilized rooms.

2. Objectives: Perform row-wise matrix calculations, conditional thresholding, and percentage formatting.

3. Flowchart:



4. Python Source Code:

```
rooms = ["C101", "C102", "C103"] occupancy = [ [1, 1, 1, 0], # C101: 3/4 = 75% [0, 1, 0, 1], # C102: 2/4 = 50% [1, 1, 1, 1] =
```

5. Test Cases & Execution Summary:

TC ID	Classroom ID	Occupancy Matrix	Utilization Rate	Status Flag
TC8.1	C101	[1, 1, 1, 0]	75.0%	HIGH UTILIZATION
TC8.2	C102	[0, 1, 0, 1]	50.0%	NORMAL

```
Room C101: Occupied 3/4 periods (75.0%) -> HIGH UTILIZATION Room C102: Occupied 2/4 periods (50.0%) ->
NORMAL Room C103: Occupied 4/4 periods (100.0%) -> HIGH UTILIZATION
```

CASE STUDY 9: STUDENT MARKS MATRIX ANALYSIS

1. Problem Statement: Process a 2D matrix storing test marks for multiple students across subjects. Compute individual student averages, subject averages, and identify the top performer.

2. Objectives: Process nested structures along row and column dimensions independently.

3. Flowchart:



4. Python Source Code:

```
students = ["Asha", "Ravi", "Noor"] subjects = ["Python", "Maths"] marks = [[88, 92], [65, 70], [95, 98]] stud_avgs = [s
```

5. Test Cases & Execution Summary:

TC ID	Metric Calculated	Expected Value	Computed Result	Status
TC9.1	Noor Average Marks	96.50	96.50	Pass
TC9.2	Top Performer	Noor	Noor (96.50)	Pass

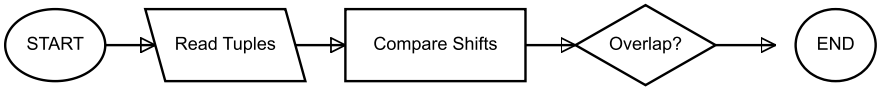
```
Student: Asha | Average: 90.00 Student: Ravi | Average: 67.50 Student: Noor | Average: 96.50 Subject:
Python | Avg: 82.67 Subject: Maths | Avg: 87.33 Top Performer: Noor (96.50)
```

CASE STUDY 10: TUPLE EMPLOYEE SHIFT CONFLICT DETECTION

1. Problem Statement: Analyze employee shift schedules formatted as tuples `(Emp_ID, Start_Hour, End_Hour)` to detect time overlap conflicts across assigned shifts.

2. Objectives: Apply tuple unpacking, range overlap verification (`startA < endB and startB < endA`), and nested record comparison.

3. Flowchart:



4. Python Source Code:

```
shifts = [("E101", 9, 14), ("E102", 12, 17), ("E103", 17, 21)] conflicts = [] for i in range(len(shifts)): for j in range(i + 1, len
```

```
Shift Conflicts Detected: [('E101', 'E102')]
```

4. PROJECT CONCLUSION & LEARNING OUTCOMES

Through the execution of these ten selected Python case studies, core principles of computational problem decomposition, state verification, string processing, and matrix manipulations were successfully implemented. The construction of ISO-compliant monochrome flowcharts ensured precise structural modeling prior to coding. Overall, this activity-based learning approach successfully built rigorous Python programming proficiency and software testing discipline.